

SAFE WORK METHOD STATEMENT (SWMS)

In order of priority:

1. **ELIMINATION** – Remove hazard completely.
2. **Substitution** – Replace object /process with less hazardous one.
3. **Isolation** – Isolate the hazard (i.e. separate the hazard from the people).
4. **Engineering** – Modify a process (i.e. redesign a workstation).
5. **Administration** – Use procedures /systems to minimise the risk (i.e. Rotation of employees, implementation of a safe work method, training, etc).
6. **Personal Protective Equipment (PPE)** – use to try to protect the person.

LIKELIHOOD	RISK SCORE CALCULATOR				
Certain to Occur	Mod.	High	High	Extreme	Extreme
Very Likely	Mod.	Mod.	High	Very High	Extreme
Possible	Low	Mod.	High	Very High	Very High
Unlikely	Low	Mod.	Mod.	High	Very High
Rare	Low	Low	Low	Mod.	High
CONSEQUENCE	LOW	MINOR	MODERATE	FATAL	CATASTROPHIC

Residual Risk	Action Required
Low	Ok for now. Continue to monitor.
MODERATE	Determine if remedial action/treatment is required.
HIGH	Immediate decision required by site leadership. Identify and commence remedial action with appropriate priority.
VERY HIGH	ACT NOW. Requires immediate attention by site leadership. Address the risk without delay. Requires urgent attention and clear, timely action plans to reduce the risk before proceeding.
EXTREME	ACT NOW. Requires immediate attention by site leadership. Urgently address the risk. Requires immediate resolution with focused attention until resolved.

Identify task high risk workplace hazards - *Non routine work is work not documented in a Safe Operating Procedure (SOP)* or Work Instruction

Note: If machine isolation is required, workers must be competent in the relevant isolation procedure.

EXTREME & VERY HIGH RISK HAZARDS	Risk	MAIN RISKS / PERMITS REQUIRED
1. Entering a Confined Space.	E	- Heat exhaustion, engulfment, limited access/egress. - Asphyxiation – low O₂ or contaminants. (Confined Space Entry Permit required) - Explosive atmosphere, combustible dusts (Confined Space Entry/Hot Work Permits required)
2. Work in an area which may have a contaminated or flammable atmosphere.	E	- Fire / explosion from ignition source. (Hot Work Permit required) - Asphyxiation or poisoning from contaminated atmosphere. (Confined Space Permit)
3. Work with potential for a fall from 2m or work on a roof	E	Falling worker/object ≥2m requires SWMS & (Risk of Falls Permit required).
4. Hot work required. (Oxy cutting, welding, soldering, grinding, etc).	E	- Fire or explosion from ignition source. (Hot Work Permit required) - Injury caused by hot surface, grinding wheel ejected pieces, welders "flash", etc. - Fire inside Expanded Polystyrene (EPS) wall panels (Hot Work Permit required).
5. Hand dig >200mm OR ground penetration >200mm OR mechanical earthworks OR enter a trench >1.5 M deep (excavating is Construction work)	E	- Contact utilities – electricity, water, gas, communications. (Excavation Permit required) - Trench Collapse. (Excavation Permit required)
6. Non routine electrical, high voltage or 'live' work or work near electrical installations.	E	- High voltage is >33kV. 'Live' work is when electrical equipment is worked on energised – (Management Approval Required) - Electrical cabinets or MCC identified as Red
7. Non routine work involving mechanical lifting, lowering, pushing or pulling.	E	- High mechanical force damage / injury. - High risk work licence for use of cranes, mobile plant, scaffold, dogging, etc.
8. Using high risk tools with excess noise, kickback, cutting edge, high speed/pressure, ejected material, weight, awkward posture, explosive force.	E	- Injury from use of high risk powered or hand tools. - Electrocution, burn or electric shock. - Manual task injury from weight, forces, overexertion, repetition or poor work heights.
9. Demolish / alter load bearing / integral part of a structure 6m high OR use load shifting machinery on a suspended floor OR use explosives. Includes modify handrail / floors.	E	- Being struck or crushed by parts of a structure or exposure to asbestos, lead, hazardous chemicals etc. (Demolition License and/or asbestos removal licence and Safe Work Plan required)
10. Work inside fences / barricades / enclosures where moving / rotating machinery is operating or hazards may exist.	V	- Serious bodily injury caused by contacting moving/unguarded plant or materials. - Machinery starting unexpectedly or electrocution, burn or electric shock.
11. Potential or stored energy which cannot be isolated (electrical, hydraulic, pneumatic, kinetic).	V	- Electrocution or entrapment. - Machinery entanglement / injury.
12. Work on or near a pressurized or enclosed circuit (water, pneumatic, hydraulic, hot oil, etc).	V	- Fracturing the mains or piping from contact – spills / leaks / emissions (environment). - Contacting temperature extremes.
13. Working in artificial temperature extremes or weather extremes. Hot, cold, wet, icy, lightning, high winds.	V	- Electrocution from using power tools / leads in wet conditions or lightning. - Heat stroke / Heat Exhaustion / Fatigue / Frost bite / Hypothermia / Influenza. - Slip / Trip / Fall Hazards.
14. Hazardous manual tasks or excessive noise or excessive vibration.	V	Personal injury/illness or hearing loss.
15. Using a hazardous chemical – chemicals, heavy metals, asbestos, Lead.	V	- Exposure to hazardous chemicals. - Spills / leaks (environmental).

CONTROLLED DOCUMENT

Doc Id: SWMS Template

Doc Owner: Tuan Safety and Compliance Officer

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Key Steps in Logical Order		Hazards/Risks to Workers or Public	Risk Score	Control Measures	Residual Risk Score
1.	Move all equipment to Hyne, Shavings shed Buckets bolts washers and nuts.	- Manual Handling - Environmental hazards - Hyne site hazards	Mod	- Team lifting using safe manual handling techniques - Hyne site induction. - Clear communication between Hyne & Tuan Biomass	Low
2.	Isolate shavings bin elevator and shavings bin discharge. Open inspection hatch	- Accidental machine startups. - Risk of dust, debris or shavings falling out when hatch is opened. - Strains or injury while opening hatch	High	- Verify the system is fully de-energised and stopped moving before starting work. - Stand clear use eye protection - Use safe manual handling techniques	Low
3.	Use impact driver to remove each bucket and replace.	-Cuts bruises or repetitive strain From accessing tight gap behind the belt. - Incorrect use of impact driver (losing control or -improper attachment of bit) - Pinch points finger may get caught while securing fasteners.	Mod	- Cut resistant gloves. - Ensure the impact driver and bits are in good condition. - All work to be carried out by competent person/s - Speciality bolts to reduce the need to access behind the belt.	Low
4.	Manually move the belt using downwards force to the next bucket	- Manual handling. - Pinch points	Mod	-Use proper lifting and pushing techniques.	Low
5.	Repeat step 3 until all buckets are replace.	-Cuts bruises or repetitive strain From accessing tight gap behind the belt. - Incorrect use of impact driver (losing control or -improper attachment of bit) - Pinch points finger may get caught while securing fasteners.	Mod	- Cut resistant gloves. - Ensure the impact driver and bits are in good condition. - All work to be carried out by competent person/s - Speciality bolts to reduce the need to access behind the belt	Low
6.	Close inspection hatch and test operation.	- Manual Handling - Pinch points finger may get caught while securing fasteners. - Accidental machine startups.	Low	- Team lifting using safe manual handling techniques. - Cut resistant gloves. - Clear communication between Test room operators and workers.	Low
8.					

Attach additional pages if required

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All relevant workers to be consulted during this process and made aware of the control measures and all required permits identified.

AUTHORISATION			
Approved by	David Knight	Position	Plant Manager
Signed		Date	

Name, Sign Off & Employer of Persons Performing Work		Start Time:	am/pm	Finish Time:	am/pm
Name	Sign Off	Employer			
1					
2					
3					

JOB COMPLETION				
1. Has equipment been modified/changed to affect how it operates?		☐ N (go to 3) ☐ Y (sign off by competent operator required)		
Has a <u>competent operator</u> approved it is safe to use? ☐ Y ☐ N	Competent Operator Name:		Competent Operator Signature:	
2. Work completed & area assessed safe before return to normal service? ☐ Y ☐ N <i>To be completed by the person conducting the task</i>		Name:		
		Signature:		
3. Altus Representative advised of & accepts changes: ☐ Y ☐ N	Name:		Signature:	